

OO Design Case Studies

1. There is an online exam to be portal, where a list of question is to be created. Every question belongs to a category and a topic. The question can be of any type such as multiple choice question, paragraph question, etc.

Design the OO model for the above problem statement and implement the code to

- Find out the total number of question
- List all question belonging to a topic
- List all question belonging to a topic and category

2. There is a training program to be conducted by some of the training providers on various courses and topic. Each training program has its own scheduling information.

Design the OO model for the above problem statement and implement the code to

- List the training program being offered by a training provider
- List all the training program scheduled on a day
- List all the training program schedule for a particular course

3. There is a restaurant which has many of its branch at various location. The restaurant offers a wide range of menu and there is also a special menu where a discount of 30% is given. Every menu is composed of minimum 3 course category have its own items.

Design the OO model for the above problem statement and implement the code to

- List the total number of menu items
- List all the menu items for a particular course category
- List all the special discount menu

4. There is a ticketing management system where a passenger books a ticket for a train and makes the payment.

Design the OO model for the above problem statement and implement the code to

- List the total amount collected
- List all the tickets for a passenger
- List all passenger for a train

5. There is a surgeon management system where a surgeon works for many hospitals and conducts operation on patients admitted in a ward. The surgeon are of various type such as Senior, Non-senior ,etc.

Design the OO model for the above problem statement and implement the code to

- Total number of patient being operated
- List all the patient being operated by a surgeon
- List all the patient admitted to a ward.

6. There is a hotel management system where a hotel contains many type of room such as single, double room, etc. And every customer books a room. The booking can be advanced or on time based on which the rent price is decided.

Design the OO model for the above problem statement and implement the code to

- Total revenue generated for a day
- List the number of vacant rooms for each room type
- List the customer details present in a particular room

7. There is a shoe collection house where different kind of shoes belonging to different brands. A customer can buy or replace a shoe from the collection house. Design the OO model for the above problem statement and implement the code to

- Number of shoe for each of its kind
- List the history of a customer transaction within a house
- List all the customer who has bought a particular shoe

8. There is a cricket tournament scheduled in the next week. Many team have enrolled and each contains minimum of 13 players. There is a scheduled fixed for each teams. Design the OO model for the above problem statement and implement the code to

- The total number of team participating in the tournament.
- List the fixture of a particular team
- List the team players for a particular team

9. A fest is going to happen in the next week. Many teams or individuals will be enrolling themselves in a one or more event by paying the amount for each event. Design the OO model for the above problem statement and implement the code to

- Total amount been collected.
- List the number of participants for each event
- List the amount collected for each event

10. A marine company has many ships with them. Ship are either cruise type, or cargo type. Some register to a particular ship to travel from one place to another or to send the cargo from one place to another. Design the OO model for the above problem statement and implement the code to

- Total amount been collected.
- List the total amount for every ship
- List all the customer details for a particular cruise ship.